

Fazleem Baig | Data Quality & Analytics Lead

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Phone: +1 902-318-3535 | Halifax, NS

Senior Data Analyst with deep technical expertise in Python, SQL, Power BI, and Databricks, and a track record of designing data pipelines and governance models that improve accuracy, compliance, and analytics speed. Experienced in leading data modernization projects for both financial and public-sector organizations, including implementing data lineage and quality frameworks using Collibra and Azure. Combines hands-on engineering skills with strategic leadership to deliver actionable insights and trusted enterprise data products.

TECHNOLOGIES

Programming Language: Python 3.x

DB & Data Warehousing: MySQL, PostgreSQL, MS SQL Server, Big Query, Hive, dbt (data build tool)

Data Visualization & BI Tools: Tableau, Looker, PowerBI

Machine Learning & Deep Learning: Scikit-learn, Tensor Flow, Keras, PyTorch, Pytorch-Lightning

Cloud Services: AWS, GCP, Azure

Tools: MS Office (Excel, Powerpoint, Word), Confluence, JIRA

Version Control: GitHub, Bitbucket

WORK EXPERIENCE

Data Quality Lead | Government of Nova Scotia, Halifax, NS | October 2024 to Present

- Directed the province's Data Quality Excellence Framework, establishing measurable standards, KPIs, and accountability models to strengthen data trust, compliance, and governance maturity across health programs.
- Led the modernization of provincial health data infrastructure, leading the migration from legacy SQL Server systems to Azure Databricks, improving data integration, scalability, and reporting efficiency.
- Operationalized enterprise-wide data quality monitoring through Power BI and Databricks dashboards, providing executives with real-time visibility into data completeness, consistency, and lineage.
- Led cross-department collaboration among Health Analytics, Finance, and Information Services teams to align governance policies and ensure CIHI-compliant reporting and standardized data definitions.
- Implemented automated Python and SQL validation pipelines, improving accuracy, reducing manual data review cycles, and accelerating readiness for analytics and data product delivery.

Senior Data Analyst | RBC Capital Markets, Halifax, NS | July 2022 to September 2024

- Orchestrated a transformative data governance initiative, enhancing data integrity and reducing error rates by 25% across key financial systems.
- Played a key role in developing and executing a comprehensive data governance strategy, focusing on data quality, privacy, and compliance using Collibra resulting in 20% increase in data lineage and management.
- Led and mentored a dynamic team of analysts, fostering a culture of continuous improvement and strategic data analysis, which contributed to a 15% increase in team productivity.
- Revolutionized reporting processes with advanced SQL and Tableau, driving a leap in analytical efficiency and report generation speed.
- Developed POCs to integrate AI and machine learning, contributing to improved operational efficiency and enriched predictive insights.

Machine Learning Solution Developer | JackHabbit, Halifax, NS | July 2021 to May 2022

- Collaborated with the product development team, Developers, and the CTO to innovate and implement cutting-edge applications, enhancing product capabilities and user experience.
- Contributed to engineering and maintenance of the backend for a growth management platform, effectively supporting a substantial userbase and enduring scalable performance under increasing demand.
- Spearheaded the design and execution of ETL pipelines, data modeling, and schema development, tailoring solutions to meet client and end-user requirements effectively.

- Developed and enforced rigorous testing protocols compliant with privacy standards, achieving an 4% improvement in system efficiency and reliability.
- Played a pivotal role in working the product development team, contributing significantly to the research and design of an MLOps strategy, thereby streamlining machine learning deployment and lifecycle management.

Data Scientist | STFX, Antigonish, NS (Contract Position) | April 2020 to September 2020

- Led the development of biomedical prediction software, enhancing accuracy in patient measure forecasts by 20% and significantly aiding medical decision-making.
- Innovated in predictive analytics by accurately projecting sparsely sampled measures (1-2 readings/hour) from dense, irregularly sampled data (1-2 Hz).
- Conducted Neural Architecture Search, using evolutionary principles to automatically create, modify, and refine neural networks, enhancing model performance and efficiency.
- Ensured 100% compliance with data protection regulations, upholding the highest standards of data security and integrity in handling confidential medical data.
- Collaborated with software development and testing teams to create scalable healthcare solutions, improving patient care quality and research efficiency.

Python Developer | TATA Consultancy Services Ltd, Bengaluru, India | July 2016 to September 2019

- Transitioned from Systems Engineer to Python Developer, identifying system enhancements and building a Django-based Automation Web Tools that reduced client's design check access time from 6 hours to 1 hour.
- Developed a secure data analysis tool for efficient engine model visualization and management, and collaborated closely with the quality assurance team to ensure software development accuracy.
- Progressed to a Data Scientist role, where I built and deployed a classification model for an insurance company, identifying fraudulent claims reducing manual effort by 30% and response time by 6 days.
- Managed the full data pipeline, including data ingestion, pre-processing, clustering, model selection, and hyperparameter tuning, using machine learning algorithms like Random Forest, SVM, and K-Means Clustering.
- Played a key role in deploying applications to the cloud and in the development of process automation tools, significantly enhancing operational efficiency and data processing capabilities.

RESEARCH PUBLICATIONS

- **Machine Learning in Pediatric Care:** Authored a groundbreaking paper titled "Machine Learning Predicts Hemodynamic Instability in Children after cardiac surgery in Pediatric Intensive Care Unit (PICU)," featured in the Proceedings of the American Heart Association Conference, November 2020. This publication highlights the application of machine learning algorithms to predict post-surgical complications in pediatric patients, demonstrating a significant advancement in critical care.
- **Innovative Use of Machine Learning in Medical Predictions:** Published "Machine learning predicts blood lactate levels in high-risk children after cardiac surgery in a pediatric intensive care unit," in The Annals of Thoracic Surgery, March 2021. The paper showcases the development and application of predictive models to enhance post-operative care in high-risk pediatric patients, contributing to improved patient outcomes and care strategies.

EDUCATION

Master of Applied Computer Science | St Francis Xavier University, Antigonish, NS | Sep 2019 - April 2021

Bachelor of Technology | SASTRA University, Thanjavur, India | June 2012 to May 2016

AWARDS AND HONORS

- **One RBC Leadership Award | RBC Capital Markets, Halifax, NS | October 2023:** Recognized for exceptional Data Analysis and Lineage deliverables, demonstrating outstanding leadership and technical expertise in complex data projects.
- **Made A Difference Award | RBC Capital Markets, Halifax, NS | August 2023:** Awarded for significant contributions to successful initiatives at RBC, exemplifying diligence, foresight, teamwork, and excellence, while consistently embodying and promoting leadership values.
- **Best Model Architecture Design Award | STFX, Antigonish, NS | November 2019:** Honored with the class award for innovative and effective model architecture in Image Recognition, showcasing advanced skills in machine learning and model development.

VOLUNTEER WORK

- **Mastermind Behind Employee Appreciation Week:** Spearheaded a delightful Food Truck event, bringing smiles and a sense of camaraderie to the workplace. It was a week filled with joy, appreciation, and delicious treats
- **Innovation Advocate at Catalyst Event:** Represented my team with zeal in a Catalyst event focused on Tooling and innovation. Played a pivotal role in sparking creative ideas and fostering a culture of innovation and teamwork.
- **Heart & Soul Volunteer at Souls Harbor Mission Mart:** Dedicated time to sorting and organizing donations at Souls Harbor Mission Mart. Contributed to making a positive impact in the lives of those in need, one item at a time.
- **Innovation Challenge 2023 Participant at Dalhousie University:** Contributed to brainstorming and problem-solving, fostering innovative ideas and creative solutions.